



EXPERIMENT NO-9

Realization of half adder using a 4 to 1 multiplexer.



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Theory:

An adder is a digital circuit that performs addition of binary numbers. The half adder adds two single binary digits A and B. It has two outputs, sum (S) and carry (C). The carry signal represents an overflow into the next digit of a multi-digit addition.

Truth Table of a Half Adder is shown in the Table (9.1).

Logical Inputs		Logical outputs	
B	A	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Table (9.1), Truth table of Half Adder.

The Boolean Expressions for Sum and Carry are;

$$\text{Sum} = \bar{B}.A + B.\bar{A} \dots\dots\dots(1)$$

$$\text{Carry} = B.A \dots\dots\dots(2)$$

To implement the half adder using 4 to 1 multiplexer we need to first understand the truth table/functional able of the multiplexer. We can see the equation shows the relation between the input and the select line of the multiplexer.

$$2^n = m \dots\dots\dots (2)$$

The block diagram of the 4 to 1 multiplexer is shown in the figure (8.1).

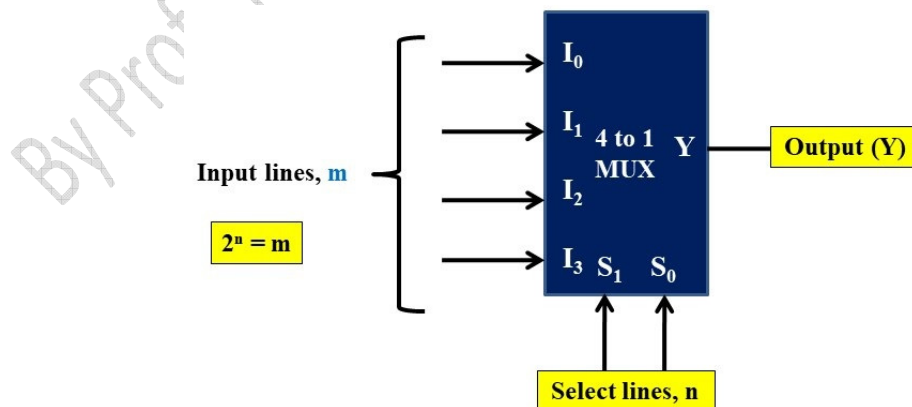


Figure (9.1), block diagram of a 4 to 1 multiplexer.

The truth table for a 4-to-1 multiplexer is depicted in Table (9.2), where the select lines assume a crucial role by accepting digital inputs and determining the chosen input directed to the output.

When the select line carries a logic zero, the output aligns with input I_0 . Likewise, the functional table in Table (8.2) succinctly outlines the output's selection based on various combinations of select lines.

Logical Select Inputs		Logical outputs
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Table 9.2, The truth table / functional table of a 4 to 1 multiplexer.

The characteristic expression can be derived from the functional table (8.1).

Since, the outputs are given as the input lines to the

$$Y = I_0 \cdot \overline{S_1} \cdot \overline{S_0} + I_1 \cdot \overline{S_1} \cdot S_0 + I_2 \cdot S_1 \cdot \overline{S_0} + I_3 \cdot S_1 \cdot S_0 \dots\dots\dots (3)$$

Let us now consider the equation (1) and write it below;

$$\text{Sum} = \overline{B} \cdot A + B \cdot \overline{A}$$

We can now expand the above equation like as shown below;

$\text{Sum} = \overline{B} \cdot \overline{A} + \overline{B} \cdot A + B \cdot \overline{A} + B \cdot A$, expanded with the missing min-terms which are in fact absent in the original expression.

$$\text{Sum} = 0 \cdot (\overline{B} \cdot \overline{A}) + 1 \cdot (\overline{B} \cdot A) + 1 \cdot (B \cdot \overline{A}) + 0 \cdot (B \cdot A) \dots\dots\dots (4)$$

We can now compare the equation (3) and (4). Let us now assume that the select line S_1 is assigned to variable B and select line S_0 as A. Then, if we compare the Inputs, we can see that we need to connect the inputs I_0 and I_3 to low logic – 0. Similarly, we can connect the inputs I_1 and I_2 to logic high.

Let us now try to realize the Carry out using the 4 to 1 multiplexer. To, realize that we need to consider the equation (2).

$$\text{Carry} = B \cdot A \dots\dots\dots (5)$$

Now, let us expand the equation (5) with the min-term which are missing.

$$\text{Carry} = 0 \cdot (\overline{B} \cdot \overline{A}) + 0 \cdot (\overline{B} \cdot A) + 0 \cdot (B \cdot \overline{A}) + 1 \cdot (B \cdot A) \dots\dots\dots (6)$$

We can now compare the equation (5) and (6). Let us now assume that the select line S_1 is assigned to variable B and select line S_0 as A. Then, if we compare the Inputs, we can see that we need to connect the inputs I_0 to I_2 to low logic – 0. Similarly, we can connect the inputs I_3 to logic high. So, to realize the carry of half adder using a 4 to 1 multiplexer we will use the IC 74153 which is a dual 4 to 1 multiplexer. One of the multiplexers will be used to realize SUM and the other to realize CARRY.

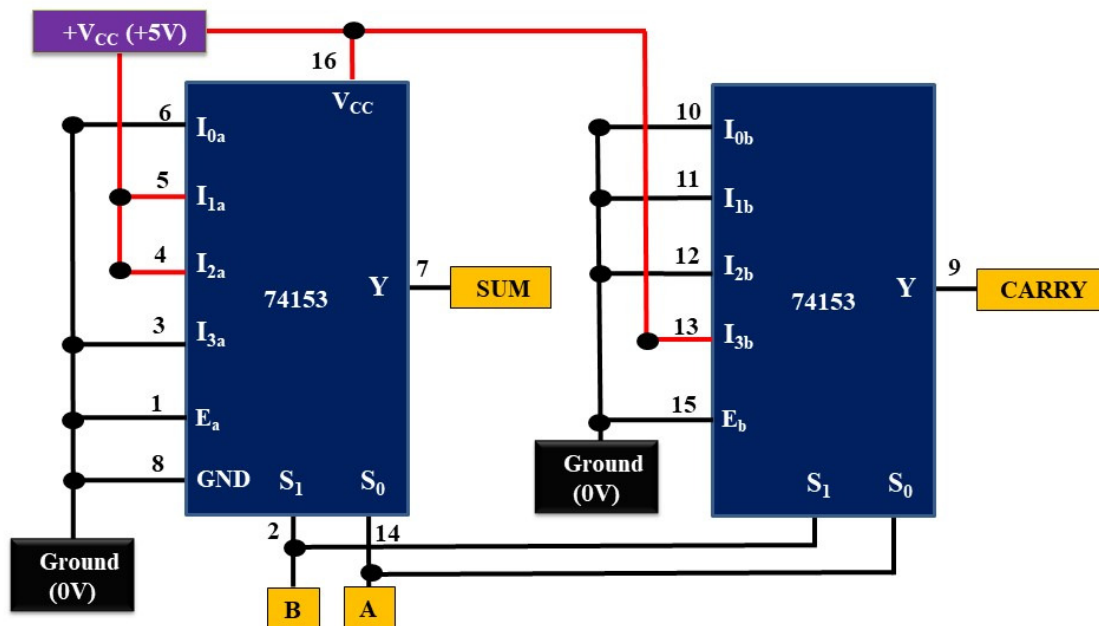


Figure (9.2), Realization of SUM and CARRY of Half adder using 4 to 1 multiplexer.

Semiconductors and Instruments required

Power Supply

1. DC +5V regulated power supply.

Integrated Circuits

1. Dual 4 to 1 multiplexer - 74153

Measuring Scientific equipment's

1. Digital Multimeter

Miscellaneous

1. Wish Board / Bread board
2. Jumper / connecting wires

Result/Observation

Truth table for the 4 to 1 multiplexer using fundamental / basic and other logic gates.

Input		Logical Output		Output Voltage	
B	A	SUM	CARRY	SUM	CARRY
0	0	0	0		
0	1	1	0		
1	0	1	0		
1	1	0	1		

Table (9.3), Experimental observation of realization of half adder using 4 to 1 multiplexer.

Discussion

1. The power supply used must be regulated to +5V dc.
2. The jumpers used on the bread board must be short.
3. All the IC used must be properly seated on the bread board along with proper connection with regulated power supply.
4. The inputs to be connected to logic - 0 must not be left floating. For logic-1 and logic-0 we need to connect the inputs either to high voltage i.e. **+5V** (V_{CC}) or to the ground potential.
5. Outputs can be visualized by connecting LED. Those must be connected via current limiting resistor.
6. In this circuit we are realizing a half adder using a dual 4 to 1 multiplexer 74153.
7. To enable the multiplexer, we need to ground or connect the pin number 1 and 15.
8. The select lines S1 and S0 are assumed to be the half adder inputs A and B. Both the multiplexer inside the 74153 has common select lines.
9. We must connect a DC regulated power supply, outputting +5 volts, to pin 16 and ground to pin 8 in order to provide the necessary supply voltage for the circuit.